Hamza Mehmood

Software Engineer | Backend, Cloud and AI

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PROFESSIONAL SUMMARY

Software engineer with 4+ years building and running production backends in C#/.NET and Python on Azure. Work includes REST APIs, event-driven services that handle retries and duplicates safely, and SQL that stays fast under load, plus Angular on the front end. Recent focus is AI: a production RAG chatbot that answers in under 3 seconds and costs 30% less to run, tool-calling agents, MCP integrations and LLM evaluation, with Claude Code, Codex and GitHub Copilot as daily tools. Earlier, cut a reporting module from 180 to 32 seconds and saw production defects fall 91%. M.S. in Computer & Information Sciences, Texas Tech University.

TECHNICAL SKILLS

Languages: C#, VB.NET, SQL, Python (Django, Flask), TypeScript, JavaScript, Java, C++

Backend and APIs: .NET 8, .NET Core, ASP.NET Core, ASP.NET MVC, Web API, Entity Framework Core, ADO.NET, REST APIs, SOAP/WCF, Microservices, Event-Driven Architecture, OAuth 2.0, JWT, Swagger/OpenAPI, Payment Processing

Cloud and DevOps: Microsoft Azure (Functions and Function Apps incl. Durable Functions, Logic Apps, API Management (APIM), Service Bus, App Service, WebJobs, Key Vault, Application Insights, Entra ID, App Configuration), Docker, Terraform, Azure DevOps, GitHub Actions, Jenkins, CI/CD, IIS

Databases: SQL Server (SSMS), Azure SQL, PostgreSQL, Oracle (PL/SQL, ODBC), Cosmos DB, Stored Procedures, Data Modeling, Schema Design, Indexing, Query Optimization, Execution Plan Analysis, Power BI

Testing and Practices: xUnit, MSTest, Postman, Test Automation, Unit Testing, Integration Testing, UML, Clean Architecture, Code Review, Agile/Scrum, Git (CLI), GitHub, SVN, Visual Studio, Technical Documentation

Frontend: WPF (MVVM), Telerik UI for WPF, Angular, TypeScript, HTML5, CSS3, Blazor

AI and LLM: LLM APIs (OpenAI, Azure OpenAI), Azure AI Foundry, LangChain, OpenRouter, MCP, RAG, Vector Databases, Prompt Engineering, LLM-as-a-Judge Evaluation, AI-Assisted Dev (Claude Code, Codex, Copilot, ChatGPT, Perplexity)

PROFESSIONAL EXPERIENCE

Research Assistant

Texas Tech University, Department of Computer Science, Lubbock, TX | Aug 2026 - Present

- Study how automated feedback generation holds up, comparing a single prompt against multi-agent LLM setups across programming, essay and mathematics tasks.
- Build LLM-as-a-judge evaluation frameworks that grade generated output against rubrics for precision, correctness and consistency.
- Run controlled Python experiments on open-source models (DeepSeek R1, Qwen), keeping run configurations and results reproducible.

Graduate Assistant, Business Technology Support

Texas Tech University - TTU Online, Lubbock, TX | Feb 2026 - Aug 2026

- Shipped a production LLM chatbot on a vector database after the first design, which sent full context every time, proved unscalable. Response time dropped to under 3 seconds and API costs fell 30%.
- Built two agents for live workflows: a Flask-based Twilio messaging agent (sending, replies, opt-out suppression, delivery tracking) and an agent that monitors content changes. Also built an AI token-consumption tracker.
- Built an evaluation pipeline that pairs a third-party API with a Python NLP analyzer, improving output-accuracy validation by 20% on real conversation logs.
- Wrote a Python data pipeline with mapping and validation logic to migrate records for 40 school districts, cutting manual effort by 60%.

Consultant .NET
Systems Limited (PartnerLinQ), Lahore, Pakistan | May 2024 - Jul 2025

- Primary production responder for a platform connecting four downstream systems: handled 30-40 incidents a month at 70% SLA compliance and cut recurring failures by 30%.
- Built event-driven services on Azure Functions, Logic Apps, API Management and Service Bus queues, with retry, idempotency and dead-letter handling built in.
- Managed production configuration with Terraform, and packaged and deployed services with Docker through Azure DevOps pipelines.
- Set up Datadog monitors and alerts and used Azure Application Insights to track failures across services. The logging and telemetry added along the way traced unresponsive apps, missing records and reconciliation mismatches back to their root cause.
- Implemented OAuth 2.0 and Azure Key Vault secrets management, and wrote about 9 XML/XSD schema mappings a month between partner systems.

Software Engineer
NETSOL Technologies, Lahore, Pakistan | Jun 2022 - Apr 2024

- Delivered 10+ modules on an enterprise lease and finance platform serving three clients, owning each from requirements through production release.
- Cut a reporting module from 180 seconds to 32 by profiling execution plans, removing table scans across two large joins, and rewriting the stored procedures with indexed views.
- Led code reviews and mentored junior developers. Production defects fell 91% and sprint delivery held at 94%.
- Worked across ASP.NET MVC and WPF with Telerik controls: built a WPF/MVVM banking channel module over WCF services, plus Angular data grids and workflow forms that client operations staff used daily.

.NET Core Developer
Hawk Logix, Lahore, Pakistan | Oct 2021 - Apr 2022

- Built 80+ REST endpoints in Clean Architecture for a US medical billing platform covering patient records, claims, billing and insurance workflows, about 30K records a month.

PROJECTS

- **OrderFlow, Event-Driven Order Processing** (C#, ASP.NET Core, EF Core) | github.com/hamzamehmood46/orderflow-dotnet: idempotent API, exponential-backoff retries and dead-letter handling; 8 concurrent duplicate requests create exactly one order.
- **Transactional Outbox and Idempotent Inbox** (C#, EF Core) | github.com/hamzamehmood46/outbox-pattern-dotnet: atomic event publishing with competing dispatchers, crash recovery and exactly-once handler execution.
- **Claims API, Medical Billing** (C#, ASP.NET Core, JWT, Clean Architecture) | github.com/hamzamehmood46/claims-api-dotnet: role-based claim workflow with a domain-enforced state machine.
- **Event-Sourced Ledger with CQRS** (C#, ADO.NET) | github.com/hamzamehmood46/event-sourced-ledger-dotnet: optimistic concurrency; parallel withdrawals can never overdraw an account.
- **AI and LLM Projects** (Python) | github.com/hamzamehmood46/WriteDetect: LLM-as-a-judge lecture scoring, AI-vs-human text detector (live demo).

EDUCATION

Master of Science, Computer & Information Sciences
Texas Tech University, Lubbock, TX | Aug 2026 | GPA 3.3/4.0

Bachelor of Science, Software Engineering
National Textile University, Pakistan | 2021 | GPA 3.5/4.0

CERTIFICATIONS

IBM Back-End Developer Professional Certificate, Coursera, 2023